

Home-Field Advantage in the 2024 SEC Football Season

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Introduction

This project will be aimed to analyze whether home-field advantage has a measurable effect on the game outcomes during the 2024 SEC football season. The goal is to determine whether SEC home teams won more often, scored more points, or showed statistically significant differences in performance compared to away teams. This project includes data cleaning, exploratory analysis, statistical tests, and future improvement.

In this analysis, these questions will be answered through statistical testing:

- Did SEC home teams have a significantly higher win percentage in 2024?
- Was the average point differential (home points - away points) significantly greater than zero?
- How did home-field advantage vary between teams within the SEC?

The dataset is saved to a local CSV that was obtained from CollegeFootballData.com using an API. The original dataset was then filtered to contain only variables needed for this analysis. The final dataset contains 53 observations and includes variables such as home team, away team, game date, game location, and final scores. The dataset allows analysis of home-field advantage within the 2024 SEC season using either `home_win` or `point_diff`.

Understanding whether home-field advantage exists helps evaluate team performance and can provide insight into the role of the environment, stadium effects, and in-game dynamics.

Results

Before making visualizations of the data, basic descriptive statistics were calculated for important variables: *point_diff* and *home_win*.

Point differential measures how many more (or fewer) points the home team scored compared to the away team in each game. Positive values indicate that the home team scored more points; negative values indicate the home team scored fewer.

Table 1: Summary statistics for point differentials in SEC home games

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-33.000	-10.000	5.000	2.057	14.000	34.000

In the summary output above, point differentials ranged from -33 to 34, indicating substantial differences for each home team in the SEC. A median of 5 and a mean of about 2 indicates that home teams typically scored slightly more than away teams, though several large losses pulled the average downward.

The variable `home_win` is a binary indicator of game outcomes for the home team: 1 if the home team won the game, and 0 if the home team lost. It shows how often SEC teams won when playing at home during the 2024 season.

Table 2: Frequency of SEC home wins in 2024

0	1
21	32

In the table above, home teams won 32 out of 53 games (about 60%), while they lost 21 games. This shows that home teams in the SEC won more often than they lost during the 2024 season.

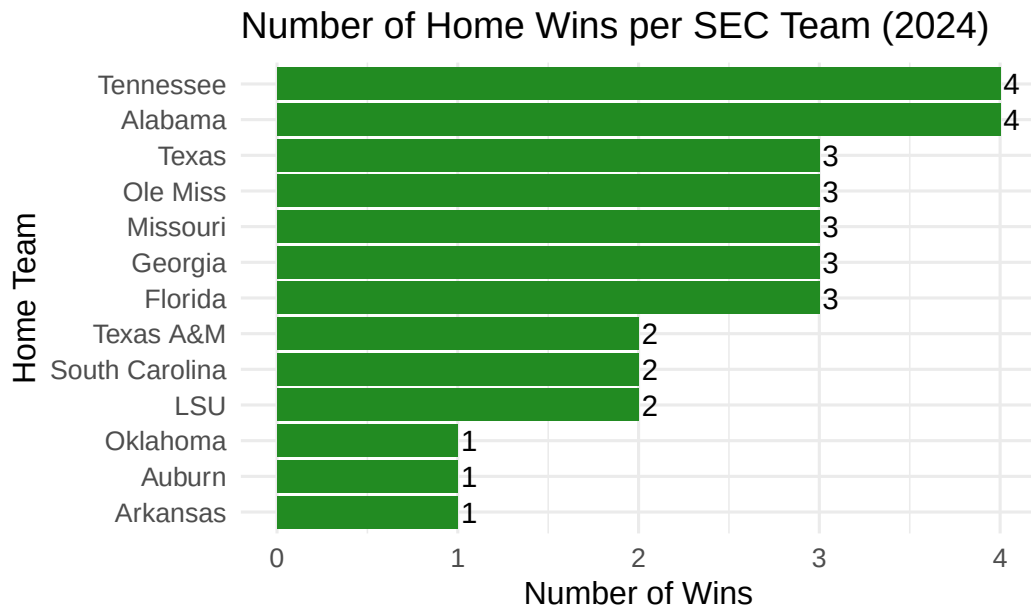


Figure 1

Summary Figure 1 displays the number of home wins recorded by each SEC team during the 2024 season. Teams are ordered from most to least home victories to make patterns easier to identify. Overall, home wins varied considerably across teams, with several teams securing multiple home victories while others won only once or not at all. This descriptive summary provides the foundation for evaluating whether home-field advantage was meaningful across the conference.

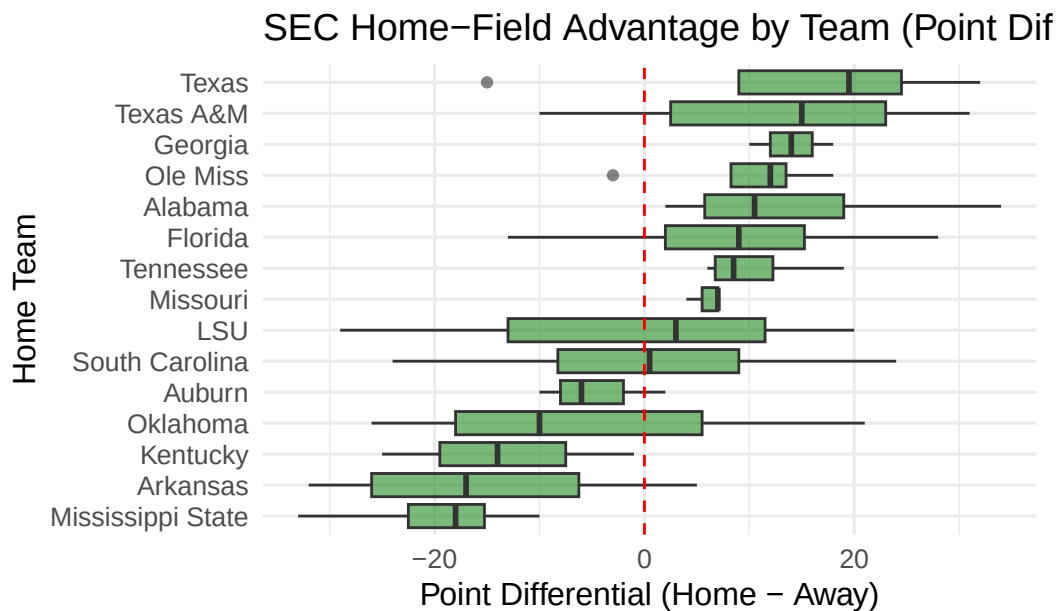


Figure 2

Summary Figure 2 shows the distribution of point differential (home points — away points) for each SEC team using boxplots. The dashed horizontal line at zero represents no advantage. Teams whose medians or distributions fall substantially above zero tend to outperform opponents at home, while teams centered near or below zero show little to no home-field advantage. This visualization highlights variability between teams, suggesting that home-field effects may depend on the specific team.

Hypothesis Tests

1. Test of Home Win Proportion

- $H_0 : p = 0.50$
- $H_1 : p \neq 0.50$

SEC home teams won **60.4%** of games in 2024. A one-sample proportion test evaluated whether this proportion was significantly greater than 50%, the expected value if home-field advantage did not exist. The result was **not statistically significant** at the 0.05 level ($p = 0.065$). The 95% confidence interval for the true proportion (0.491–1.000) includes 0.50.

Conclusion: There is no statistically significant evidence that SEC home teams won more often than expected by chance, although the result is borderline.

2. Test of Mean Point Differential

- $H_0: \mu = 0$
- $H_1: \mu \neq 0$

The average point differential was **+2.06 points**, meaning home teams scored slightly more than away teams on average. A one-sample t-test assessed whether this mean was significantly greater than zero. The result was **not significant** ($t = 0.86$, $p = 0.196$). The 95% one-sided confidence interval ($-1.94, \infty$) includes zero.

Conclusion: SEC home teams did not outscore away teams by a statistically meaningful margin in 2024.

3. Differences in Home-Field Advantage Across Teams (ANOVA)

A one-way ANOVA tested whether the average point differential differed among SEC teams. The effect of **home_team** was **statistically significant** ($F = 2.06$, $p = 0.039$), indicating that some teams performed better at home than others.

Conclusion: Home-field advantage was not uniform across the conference.

4. Tukey Post-Hoc Analysis

A Tukey HSD test examined which specific teams differed from each other. Although the ANOVA showed an overall effect, **no pairwise comparisons were statistically significant** once multiple-comparison adjustments were applied (all adjusted p-values > 0.05).

Conclusion: There is collective evidence of team-to-team variability, but no individual team displayed a significantly stronger or weaker home-field advantage relative to another.

Discussion

Summary of Findings

This analysis evaluated home-field advantage in the 2024 SEC football season through summary statistics, visualizations, and hypothesis testing. Although home teams won 60.4% of games, neither the win proportion nor the average point differential provided statistically significant evidence of a conference-wide home-field advantage. However, the ANOVA results indicate that some teams may benefit more from playing at home than others, even if no individual pairwise differences reached significance in post-hoc testing.

Overall, the findings suggest that any home-field advantage in the 2024 SEC season was slight and inconsistent, varying by team rather than being a universal factor.

Limitations

Several limitations should be considered when interpreting these findings:

- **Small sample size:** Only 53 SEC vs. SEC games were included, reducing statistical power.
- **Single-season analysis:** Results may reflect unusual patterns specific to the 2024 season.
- **Uncontrolled variables:** Factors such as stadium noise, team strength, injuries, weather, and travel distance were not accounted for.

These limitations may hide subtle but real home-field effects.

Future Directions

If this analysis were expanded or repeated, several enhancements could be made like including multiple seasons to improve power and identify trends over time, and incorporating additional variables, such as crowd noise, attendance, penalties, or pre-game betting spreads. Also, evaluating non-conference games to broaden the dataset and test whether SEC teams behave differently at home against weaker or stronger opponents. Another direction could be to analyze situational factors, such as rivalry games, night vs. day games, or weather.

Future research could also explore how environmental factors, especially stadium noise levels, influence away team performance.